1.(1) $\frac{e-2.7}{e}$

$|x-x^*| < 0.02 < \frac{1}{2}\times 10^{-1} = \frac{1}{2}\times 10^{1-2}$ 2位

(2) $\frac{e-2.718}{e}$

$|x-x^*| < 0.0003 < \frac{1}{2}\times 10^{-3} = \frac{1}{2}\times 10^{1-4}$ 4位

(3) ~~$\frac{e-2.7}{e}$~~ $\frac{\frac{e}{100}-0.027}{\frac{e}{100}} = \frac{e-2.7}{e}$

$|x-x^*| < 0.0002 < \frac{1}{2}\times 10^{-3} = \frac{1}{2}\times 10^{1-2}$ 2位

(4) $\frac{\frac{e}{100}-0.02718}{\frac{e}{100}} = \frac{e-2.718}{e}$

$|x-x^*| < 0.00000\,3 < \frac{1}{2}\times 10^{-5} = \frac{1}{2}\times 10^{1-4}$ 4位

2. (1) $\int_N^{N+1} \frac{1}{1+x^2}\,dx = \arctan(N+1) - \arctan N$

$= \arctan \frac{1}{N^2+N+1}$

$= \frac{1}{N^2+N+1} - \frac{1}{3(N^2+N+1)^3} + \frac{1}{5(N^2+N+1)^5} \approx \frac{1}{N^2+N+1}$

$\sum_{n=0}^{\infty} \frac{(-1)^n \left(\frac{1}{N^2+N+1}\right)^{2n+1}}{2n+1}$

(2) $\sqrt{x+\frac{1}{x}} - \sqrt{x-\frac{1}{x}} = \frac{2}{x\left(\sqrt{x+\frac{1}{x}}+\sqrt{x-\frac{1}{x}}\right)} = \frac{2}{x\sqrt{x}\left(\sqrt{1+\frac{1}{x^2}}+\sqrt{1-\frac{1}{x^2}}\right)}$

$= \frac{2}{x\sqrt{x}\left(1+\frac{1}{2x^2}+o\left(\frac{1}{x^2}\right)+1-\frac{1}{2x^2}+o\left(\frac{1}{x^2}\right)\right)}$

$= \frac{2}{x\sqrt{x}\left(2+o\left(\frac{1}{x^2}\right)\right)}$

$= \frac{1}{x\sqrt{x}} \approx \frac{1}{x\sqrt{x}}$

$= \frac{2}{x\sqrt{x}\left(2+\sum_{n=2}^{\infty}\frac{(-1)^{n-1}(2n-3)!!}{2^n\cdot n!}\left(\frac{1}{x^2}\right)^n + \sum_{n=2}^{\infty}\frac{(-1)^{n-1}(2n-3)!!}{2^n\cdot n!}\left(-\frac{1}{x^2}\right)^n\right)}$

$= \frac{1}{x\sqrt{x}\left(1+\sum_{k=1}^{\infty}\frac{-(2k-3)!!}{2^{2k}\cdot(2k)!}\left(\frac{1}{x^2}\right)^{2k}\right)} \approx \frac{1}{x\sqrt{x}}$

(3) $\ln(x+1)-\ln x = \ln\left(1+\frac{1}{x}\right)$

$= \frac{1}{x} - \frac{1}{2}\left(\frac{1}{x}\right)^2 \sum_{n=0}^{\infty}\frac{(-1)^n\left(\frac{1}{x}\right)^{n+1}}{n+1} = \sum_{n=0}^{\infty}\frac{(-1)^n}{(n+1)x^{n+1}} \approx \frac{1}{x}$

(4) $\cos^2 x - \sin^2 x = (\cos x+\sin x)(\cos x-\sin x) = -2\cos\left(x-\frac{\pi}{4}\right)\cdot\sin\left(x-\frac{\pi}{4}\right)$

$= -2\left[\sum_{n=0}^{\infty}\frac{(-1)^n\left(x-\frac{\pi}{4}\right)^{2n}}{(2n)!}\right]\cdot\left[\sum_{n=0}^{+\infty}\frac{(-1)^n\left(x-\frac{\pi}{4}\right)^{2n+1}}{(2n+1)!}\right] \approx \frac{\pi}{2}-2x$